Question ID: 84664a7c

Question

The front of a roller-coaster car is at the bottom of a hill and is 15 feet above the ground. If the front of the roller-coaster car rises at a constant rate of 8 feet per second, which of the following equations gives the height h, in feet, of the front of the roller-coaster car s seconds after it starts up the hill?

Answer

A. $h = 8s + 15$

B. $h = 15s + \frac{335}{8}$

C. $h = 8s + \frac{335}{15}$

D. $h = 15s + 8$

Question ID: 7fac16fb

Question

The function f is defined by $f(x) = \frac{7}{10}x + 55$. What is the value of $f(20)$?

Question ID: 06fc1726

Question

If f is the function defined by $f(x) = \frac{2x-1}{3}$, what is the value of $f(5)$?

Answer

A. $\frac{4}{3}$

B. $\frac{7}{3}$

C. 3

D. 9

Question ID: 0eae6be1

Question

The number y is 84 less than the number x. Which equation represents the relationship between x and y?

Answer

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

Question ID: bf36c815

Question

The function g is defined by $g(x) = -x + 8$. What is the value of $g(0)$?

Answer

A. -8

B. 0

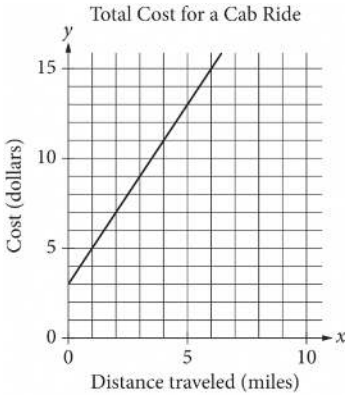
C. 4

D. 8

Question ID: 3f5375d9

Question

The line graphed in the xy-plane below models the total cost, in dollars, for a cab ride, y, in a certain city during nonpeak hours based on the number of miles traveled, x.



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

Answer

A. $2.00

B. $2.60

C. $3.00

D. $5.00

Question ID: 12983c1e

Question

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above. Which of the following defines f ?

Answer

A. $f(x)=2x+3$

B. $f(x)=3x+2$

C. $f(x)=4x+1$

D. $f(x)=5x$

Question ID: a396ed75

Question

For a training program, Juan rides his bike at an average rate of 5.7 minutes per mile. Which function m models the number of minutes it will take Juan to ride x miles at this rate?

Answer

A. $m(x) = \frac{x}{5.7}$

B. $m(x) = x + 5.7$

C. $m(x) = x - 5.7$

D. $m(x) = 5.7x$

Question ID: 3462d850

Question

Marisol drove 3 hours from City A to City B. The equation below estimates the distance d, in miles, Marisol traveled after driving for t hours.

$$d = 45t$$

Which of the following does 45 represent in the equation?

Answer

A. Marisol took 45 trips from City A to City B.

B. The distance between City A and City B is 45 miles.

C. Marisol drove at an average speed of about 45 miles per hour.

D. It took Marisol 45 hours to drive from City A to City B.

Question ID: 9d9fe1e6

Question

In science class, Diego conducted an experiment to learn about evaporation. Diego measured the height of fluid in a beaker over a period of time. The function $f(x) = 39 - 0.18x$ gives the estimated height, $\textbf{in centimeters } (\text{cm})$, of the fluid in the beaker x days after the start of the experiment. Which of the following is the best interpretation of 39 in this context?

Answer

A. The estimated height, $\textbf{in cm}$, of the fluid at the start of the experiment

B. The estimated height, $\textbf{in cm}$, of the fluid at the end of the experiment

C. The estimated change in the height, $\textbf{in cm}$, of the fluid each day

D. The estimated number of days for all the fluid to evaporate

Question ID: 255996a6

Question

$T = 1{,}000 + 18h$

In the equation above, T represents Brittany's total take-home pay, in dollars, for her first week of work, where h represents the number of hours she worked that week and 1,000 represents a sign-on bonus. If Brittany's total take-home pay was $1,576, for how many hours was Brittany paid for her first week of work?

Answer

A. 16

B. 32

C. 55

D. 88

Question ID: a1696f3e

Question

The function g is defined as $g(x) = 5x + a$, where a is a constant. If $g(4) = 31$, what is the value of a ?

Answer

A. 30

B. 22

C. 11

D. -23

Question ID: de6fe450

Question

On January 1, 2015, a city's minimum hourly wage was $9.25. It will increase by $0.50 on the first day of the year for the next 5 years. Which of the following functions best models the minimum hourly wage, in dollars, x years after January 1, 2015, where $x = 1, 2, 3, 4, 5$?

Answer

A. $f(x) = 9.25 - 0.50x$

B. $f(x) = 9.25x - 0.50$

C. $f(x) = 9.25 + 0.50x$

D. $f(x) = 9.25x + 0.50$

Question ID: 6efcc0a3

Question

In the linear function h, $h(0) = 41$ and $h(1) = 40$. Which equation defines h?

Answer

A. $h(x) = -x + 41$

B. $h(x) = -x$

C. $h(x) = -41x$

D. $h(x) = -41$

Question ID: 97eab129

Question

Area (square feet)	Water (gallons)
2,520	4,536
3,780	6,804
5,040	9,072

The buildings of a shopping center are designed to allow water to drain from the roof into gutters on the sides of the buildings. The table shows the relationship between the area x, in square feet, of a roof and the amount of water $f(x)$, in gallons, drained from the roof into the gutters over a certain period of time. Which equation could define f?

Answer

A. $f(x) = 0.6x$

B. $f(x) = 1.8x$

C. $f(x) = 2{,}268x$

D. $f(x) = 4{,}536x$

Question ID: 2eef7e61

Question

The graph of the function f is a line in the xy-plane. If the line has slope $\frac{3}{4}$ and $f(0)=3$, which of the following defines f?

Answer

A. $f(x)=\frac{3}{4}x-3$

B. $f(x)=\frac{3}{4}x+3$

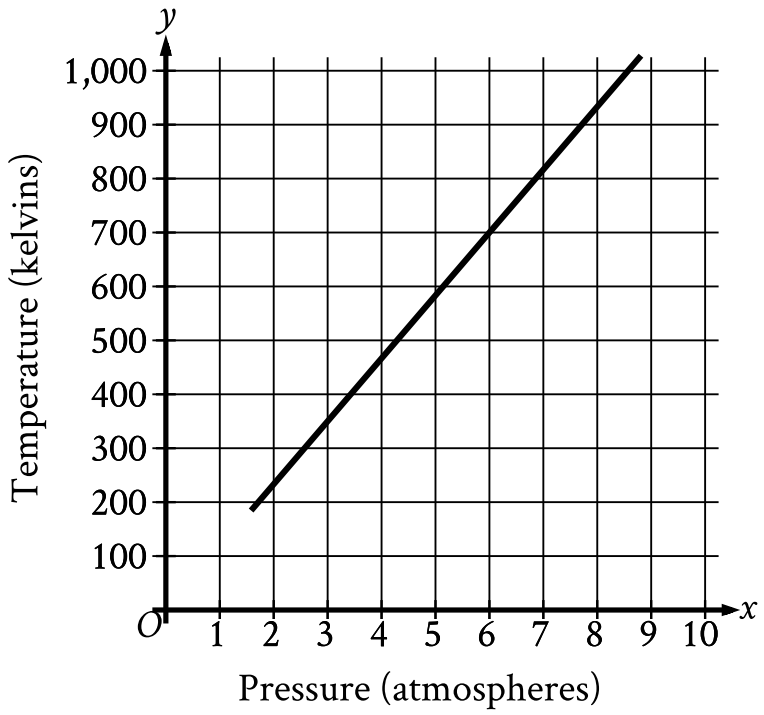
C. $f(x)=4x-3$

D. $f(x)=4x+3$

Question ID: 0ea7ef01

Question

Oxygen gas is placed inside a tank with a constant volume. The graph shows the estimated temperature y, in kelvins, of the oxygen gas when its pressure is x atmospheres.



What is the estimated temperature, in kelvins, of the oxygen gas when its pressure is 6 atmospheres?

Answer

A. 6

B. 60

C. 700

D. 760

Question ID: 1ecaa9c0

Question

Robert rented a truck to transport materials he purchased from a hardware store. He was charged an initial fee of $20.00 plus an additional $0.70 per mile driven. If the truck was driven 38 miles, what was the total amount Robert was charged?

Answer

A. $46.60

B. $52.90

C. $66.90

D. $86.50

Question ID: 8643d906

Question

$P(t) = 250 + 10t$

The population of snow leopards in a certain area can be modeled by the function P defined above, where $P(t)$ is the population t years after 1990. Of the following, which is the best interpretation of the equation $P(30) = 550$?

Answer

A. The snow leopard population in this area is predicted to be 30 in the year 2020.

B. The snow leopard population in this area is predicted to be 30 in the year 2030.

C. The snow leopard population in this area is predicted to be 550 in the year 2020.

D. The snow leopard population in this area is predicted to be 550 in the year 2030.

Question ID: a4d6fbec

Question

If $y = 5x + 10$, what is the value of y when $x = 8$?

Question ID: a73a5c22

Question

The function g is defined by $g(x) = 10x + 8$. What is the value of $g(x)$ when $x = 8$?

Answer

A. 0

B. 8

C. 10

D. 88

Question ID: 1993561d

Question

For the linear function f, the graph of $y = f(x)$ in the *xy*-plane has a slope of $\frac{1}{4}$ and passes through the point $(0, 5)$. Which equation defines f?

Answer

A. $f(x) = \frac{1}{4}x + 5$

B. $f(x) = \frac{1}{4}x + \frac{1}{5}$

C. $f(x) = \frac{1}{4}x - \frac{5}{4}$

D. $f(x) = \frac{1}{4}x - 5$

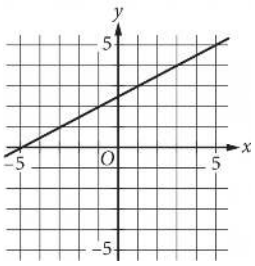
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Question

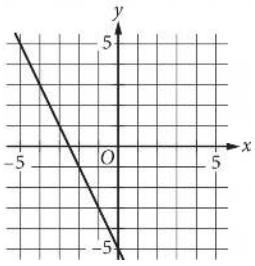
Which of the following is the graph of the equation $y = 2x - 5$ in the xy-plane?

Answer

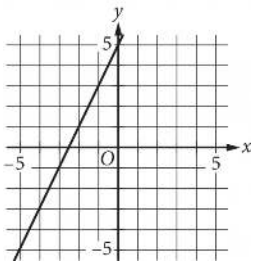
A.



B.



C.



D.

